

ESIM Bench Test

2018 Stelvio 2.0T | ESIM Mopar 04861961AD, fitted new 2026-09-02 | EVAP smoke-tested sealed 3 times

Purpose: confirm the new ESIM's switch closes under a small vacuum, its seal holds, and it vents under light pressure. Do **wiTECH Test A first** if you can: it tests the ESIM in place plus its wiring. Bench-test only if Test A fails or no ESIM parameter exists in wiTECH. Compare against a new ESIM where a number matters.

Safety: fuel vapour, no sparks or flames; ignition OFF before unplugging; apply vacuum and pressure **gently**; never shop air.
Tools: multimeter (ohms), hand vacuum pump, low-range gauge or water manometer, rubber caps.

Steps

- **Access:** remove the driver-side rear wheel liner; unplug the ESIM (ignition off). Inspect the connector: corrosion, water, bent or pushed-back pins.
- **Remove** the ESIM or free both hoses. Mark the **canister-side** and **fresh-air** ports.
- **At rest:** meter across the switch pins: expect **open**. (More than 2 pins: find the pair that changes in the next step.)
- **Vacuum** on the canister-side port, fresh-air side open: pump slowly. Switch should snap **closed** (near 0 ohms) at a small vacuum. Note the gauge reading.
- **Hold** 1 minute: vacuum steady, switch stays closed (seal test).
- **Release:** switch opens again.
- **Pressure:** gentle puff into the canister-side port: it should vent through, not hold.
- **Tap and wiggle** while held closed: no flicker.

Result

You see	Means	Next
Never closes	Failed switch: ECU sees a leak on a sealed system	Replace ESIM only (TSB 9100469)
Closes, vacuum bleeds away	Internal seal leak	Replace ESIM only
Closes only at much higher vacuum than new	Sluggish switch	Replace ESIM
Flickers when tapped	Failing contacts	Replace ESIM; check harness connector
Passes every step	ESIM good	Wiring (Test B) or ECU calibration (flash check)

Record

Item	Result
Date / connector condition	
At rest (open?)	
Vacuum where it closed	
Held 1 min? / reopened on release?	
Vents under light pressure? / flicker?	
New-unit comparison / conclusion	